Project: Worked examples
Title: EN 1993-1-1 member checks (basic)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A\, f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W\, f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v\, f_y/(\sqrt{3}\,\gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\, V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$, resistance $\times (1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\, f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\, A\, f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$
Member check, axial force and bending with buckling	$N_{Ed}/N_{b,Rd,min} + k_{yy}\, M_{y,Ed}/M_{y,Rd} + k_{zz}\, M_{z,Ed}/M_{z,Rd} \le 1$

Basis: Tension[1](6.2.3); Compression[1](6.2.4); Bending moment[1](6.2.5); Shear[1](6.2.6); Bending and shear[1](6.2.8); Linear summation of utilisations[1](6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](6.3.3); Imperfection factors for the buckling curves[2](§6.3.1).

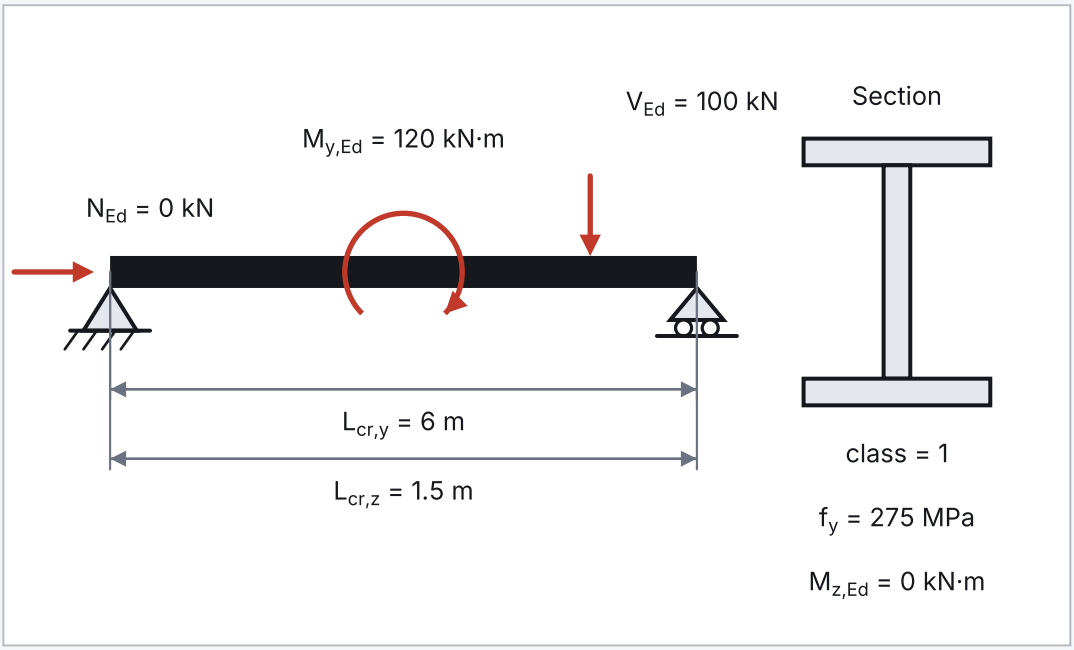


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{Ed}	Design axial force (tension +, compression −)	0	kN	reference: S2 frame sheet, two-span beam, ULS	-100000 … 100000 kN
$M_{y,Ed}$	Design moment about y-y (major)	120	kN·m	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN·m
$M_{z,Ed}$	Design moment about z-z (minor)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_{Ed}	Design shear force	100	kN	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN
f_y	Yield strength (for the thickness)	275	MPa	spec: BS EN 10025-2 S275, t ≤ 16 mm	200 … 700 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
γ_{M0}	Partial factor γ_{M0} (cross-section resistance)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M1}	Partial factor γ_{M1} (member buckling)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
$section_{class}$	Cross-section class	1		assumption: UB 305×165×40 S275 is class 1 in bending (Blue Book).	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Area A	51.3	cm^2	reference: S1 section sheet / Blue Book UB 305×165×40	1 … 100000 cm^2
$W_{pl,y}$	$W_{pl,y}$	623	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,y}$	$W_{el,y}$	560	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{pl,z}$	$W_{pl,z}$	142	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,z}$	$W_{el,z}$	92.6	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	I_y	8503	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	I_z	764	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth h (shear-area helper)	303.4	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
b	Width b (helper)	165	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
t_w	Web thickness (helper)	6	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
t_f	Flange thickness (helper)	10.2	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
r	Root radius (helper)	8.9	mm	spec: BS EN 10365 UB 305×165×40	0 … 500 mm
η	Shear-area factor η	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
$L_{cr,y}$	Buckling length about y-y	6	m	assumption: Span 6 m.	0.1 … 100 m
$L_{cr,z}$	Buckling length about z-z	1.5	m	assumption: Compression flange restrained at 1.5 m by the deck.	0.1 … 100 m
$curve_y$	Buckling curve about y-y	a		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, y-y	any
$curve_z$	Buckling curve about z-z	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, z-z	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_{yy}	Interaction factor k_{yy} (member check)	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2
k_{zz}	Interaction factor k_{zz} (member check)	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2

3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. Lateral-torsional buckling is not checked: the compression flange is assumed restrained, or LTB is checked elsewhere. (6.3.2 not implemented.)
A3. The member interaction uses the user's k_{yy} and k_{zz} (default 1.0) with the smaller flexural-buckling reduction on the axial term; the Annex A/B interaction factors are not derived. (6.3.3 simplified.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. $\text{section}_{\text{class}}$ (Cross-section class) is taken as 1. (UB 305×165×40 S275 is class 1 in bending (Blue Book).)
A6. $L_{cr,y}$ (Buckling length about y-y) is taken as 6 m. (Span 6 m.)
A7. $L_{cr,z}$ (Buckling length about z-z) is taken as 1.5 m. (Compression flange restrained at 1.5 m by the deck.)
A8. k_{yy} (Interaction factor k_{yy} (member check)) is taken as 1. (No axial force; interaction factors irrelevant.)
A9. k_{zz} (Interaction factor k_{zz} (member check)) is taken as 1. (No axial force; interaction factors irrelevant.)

4. Working

[27] $r_h = r$
$= 8.9\ \text{mm}$
$= 8.9\ \text{mm}$
Root radius used by the helper

[28] $A_{v,rolled} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ [1](6.2.6(3))
$= 51.3\ \text{cm}^2 - 2 \times 165\ \text{mm} \times 10.2\ \text{mm} + (6\ \text{mm} + 2 \times 8.9\ \text{mm}) \times 10.2\ \text{mm}$
$= 20.068\ \text{cm}^2$
Rolled I shear area, load parallel to the web

[29] $A_{v,min} = \eta * (h - 2 * t_f) * t_w$ [1](6.2.6(3))
$= 1 \times (303.4\ \text{mm} - 2 \times 10.2\ \text{mm}) \times 6\ \text{mm}$
$= 16.98\ \text{cm}^2$
Lower bound $\eta\ h_w\ t_w$

[30] $A_v = \max(A_{v,rolled}, A_{v,min})$
$= \max(20.068\ \text{cm}^2, 16.98\ \text{cm}^2)$
$= 20.068\ \text{cm}^2$
Shear area

[31] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [1](6.2.3(2), 6.2.4(2))
$= 51.3\ \text{cm}^2 \times 275\ \text{MPa} / 1$
$= 1410.8\ \text{kN}$
Plastic axial resistance

[32] $W_y = W_{pl,y}$ [1](6.2.5(2))
$= 623\ \text{cm}^3$
$= 623\ \text{cm}^3$
Section modulus about y (plastic, class 1–2)

[33] $W_z = W_{pl,z}$ [1](6.2.5(2))
= 142 cm³
= 142 cm³
Section modulus about z (plastic)
[34] $V_{pl,Rd} = A_v * f_y / (\sqrt{3} * \gamma_{M0})$ [1](6.2.6(2))
= 20.068 cm² × 275 MPa / (sqrt(3) × 1)
= 318.62 kN
Plastic shear resistance
[35] $u_V = V_{Ed} / V_{pl,Rd}$
= 100 kN / 318.62 kN
= 0.31386
Shear utilisation
[36] $\rho = 0$ [1](6.2.8(2))
= 0
Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)
[37] $M_{y,c,Rd} = (1 - \rho) * W_y * f_y / \gamma_{M0}$ [1](6.2.5(2), 6.2.8)
= (1 − 0) × 623 cm³ × 275 MPa / 1
= 171.32 kN·m
Bending resistance about y
[38] $M_{z,c,Rd} = (1 - \rho) * W_z * f_y / \gamma_{M0}$ [1](6.2.5(2))
= (1 − 0) × 142 cm³ × 275 MPa / 1
= 39.05 kN·m
Bending resistance about z
[39] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$
= 120 kN·m / 171.32 kN·m
= 0.70042
Bending utilisation, y
[40] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$
= 0 kN·m / 39.05 kN·m
= 0
Bending utilisation, z
[41] $u_N = abs(N_{Ed}) / N_{pl,Rd}$
= abs(0 kN) / 1410.8 kN
= 0
Tension utilisation
[42] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [1](6.2.1(7))
= 0 + 0.70042 + 0
= 0.70042
Linear cross-section interaction
[45] α_y = imperfection factor, curve a [1](6.3.1.2)
= Designers' Guide §6.3.1
= 0.21
Imperfection factor about y
[46] α_z = imperfection factor, curve b [1](6.3.1.2)
= Designers' Guide §6.3.1
= 0.34
Imperfection factor about z
[47] $N_{cr,y} = pi^2 * E * I_y / L_{cr,y}^2$ [1](6.3.1.2(1))
= π^2 × 210 GPa × 8503 cm⁴ / (6 m)^2
= 4895.4 kN
Elastic critical force about y
[48] $\lambda_y = \sqrt{A * f_y / N_{cr,y}}$ [1](6.3.1.2(1))
= sqrt(51.3 cm² × 275 MPa / 4895.4 kN)
= 0.53682
Non-dimensional slenderness about y

[49] $\Phi_y = 0.5 * (1 + \alpha_y * (\lambda_y - 0.2) + \lambda_y^2)$ [1](6.3.1.2(1))
$= 0.5 \times (1 + 0.21 \times (0.53682 - 0.2) + 0.53682^2)$
$= 0.67946$
Φ about y
[50] $\chi_y = \min(1, 1 / (\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}))$ [1](6.3.1.2(1))
$= \min(1, 1 / (0.67946 + \sqrt{0.67946^2 - 0.53682^2}))$
$= 0.91244$
Reduction factor χ about y
[51] $N_{b,y,Rd} = \chi_y * A * f_y / \gamma_{M1}$ [1](6.3.1.1(3))
$= 0.91244 \times 51.3\ cm^2 \times 275\ MPa / 1$
$= 1287.2\ kN$
Buckling resistance about y
[52] $N_{cr,z} = pi^2 * E * I_z / L_{cr,z}^2$ [1](6.3.1.2(1))
$= \pi^2 \times 210\ GPa \times 764\ cm^4 / (1.5\ m)^2$
$= 7037.7\ kN$
Elastic critical force about z
[53] $\lambda_z = \sqrt{A * f_y / N_{cr,z}}$ [1](6.3.1.2(1))
$= \sqrt{51.3\ cm^2 \times 275\ MPa / 7037.7\ kN}$
$= 0.44772$
Non-dimensional slenderness about z
[54] $\Phi_z = 0.5 * (1 + \alpha_z * (\lambda_z - 0.2) + \lambda_z^2)$ [1](6.3.1.2(1))
$= 0.5 \times (1 + 0.34 \times (0.44772 - 0.2) + 0.44772^2)$
$= 0.64234$
Φ about z
[55] $\chi_z = \min(1, 1 / (\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}))$ [1](6.3.1.2(1))
$= \min(1, 1 / (0.64234 + \sqrt{0.64234^2 - 0.44772^2}))$
$= 0.90667$
Reduction factor χ about z
[56] $N_{b,z,Rd} = \chi_z * A * f_y / \gamma_{M1}$ [1](6.3.1.1(3))
$= 0.90667 \times 51.3\ cm^2 \times 275\ MPa / 1$
$= 1279.1\ kN$
Buckling resistance about z
[57] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$
$= \min(1287.2\ kN, 1279.1\ kN)$
$= 1279.1\ kN$
Governing buckling resistance
[58] $u_{buckling} = 0$ [1](6.3.1.1(1))
$= 0$
Flexural buckling utilisation (tension: not applicable)
[59] $u_{member} = u_{M,y} + u_{M,z}$ [1](6.3.3(4))
$= 0.70042 + 0$
$= 0.70042$
Simplified member interaction

5. Results

SYMBOL	VALUE	DESCRIPTION
A_v	20.068 cm²	Shear area
$N_{pl,Rd}$	1410.8 kN	$N_{pl,Rd}$
$V_{pl,Rd}$	318.62 kN	$V_{pl,Rd}$
ρ	0	ρ
$M_{y,c,Rd}$	171.32 kN·m	$M_{y,c,Rd}$

SYMBOL	VALUE	DESCRIPTION
$M_{z,c,Rd}$	39.05 kN·m	$M_{z,c,Rd}$
$N_{cr,y}$	4895.4 kN	$N_{cr,y}$
$N_{cr,z}$	7037.7 kN	$N_{cr,z}$
λ_y	0.53682	λ_y
λ_z	0.44772	λ_z
χ_y	0.91244	χ_y
χ_z	0.90667	χ_z
$N_{b,y,Rd}$	1287.2 kN	$N_{b,y,Rd}$
$N_{b,z,Rd}$	1279.1 kN	$N_{b,z,Rd}$
$N_{b,Rd}$	1279.1 kN	$N_{b,Rd}$
u_N	0	Axial utilisation
$u_{M,y}$	0.70042	Bending y utilisation
$u_{M,z}$	0	Bending z utilisation
u_V	0.31386	Shear utilisation
$u_{section}$	0.70042	Cross-section interaction
$u_{buckling}$	0	Buckling utilisation
u_{member}	0.70042	Member interaction

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bending y $M_{y,Ed} \le M_{y,c,Rd}$	$u_{M,y} = 0.70042$	$u_{M,y} \le 1$	0.7	[1](6.2.5)
PASS	Cross-section interaction	$u_{section} = 0.70042$	$u_{section} \le 1$	0.7	[1](6.2.1(7))
PASS	Shear $V_{Ed} \le V_{pl,Rd}$	$u_V = 0.31386$	$u_V \le 1$	0.314	[1](6.2.6)
PASS	Tension $N_{Ed} \le N_{pl,Rd}$	$u_N = 0$	$u_N \le 1$	0	[1](6.2.3)
PASS	Bending z $M_{z,Ed} \le M_{z,c,Rd}$	$u_{M,z} = 0$	$u_{M,z} \le 1$	0	[1](6.2.5)

7. Applicability

WARNING — no-ltb: $M_{y,Ed}$: lateral-torsional buckling is not checked by this sheet; restrain the compression flange or check 6.3.2 separately.

8. Verification

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Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.